

Individual project Machine Learning for Business

Deadline: 30/03/2025, 23:59. Submit your *.py or *.ipynb files and a PDF with answers via the K2 course portal. Use of generative AI is not allowed.

Project Description

Predict house prices in different communities based on air pollution level (nitrogen oxide concentration), distance from employment centers, student-teacher ratio, average number of rooms, and crime rate.

Variables

- **price:** median housing price
- **crime:** crimes committed per capita
- **nox:** nitrogen oxide, parts per 100 million
- **rooms:** average number of rooms per house
- **dist:** weighted distance to 5 employment centers
- **stratio:** average student-teacher ratio
- **lprice:** $\log(\text{price})$
- **lnox:** $\log(\text{nox})$

Data file: hprice2_merged.csv

Tasks

1. Randomly select two sets of 100 data points. Compute mean, variance, and histogram for *lprice*, *lnox*, *rooms*, *dist*, and *crime*. Are values similar? Can we assume independent and identically distributed variables? Use hypothesis testing to compare distributions.
2. Fit linear regression models using subsets of $\{\textit{lnox}, \textit{dist}, \textit{rooms}, \textit{stratio}, \textit{crime}\}$ to predict *lprice*. Reserve at least 20% of data for validation.

3. Calculate R^2 scores for each model. Comment on model quality. Are independent variables linearly independent? Which variables are most relevant for predicting $lprice$ and why?
4. Evaluate confidence intervals for regression parameters for each model. Are all coefficients statistically significant? Explain.
5. Fit a model using $lnox$, $dist$, $rooms$, $stratio$, and $\log(nox^3e^{dist+1})$ to predict $lprice$. Compute confidence intervals. Compare variable significance with Step 2. Is there multicollinearity?
6. Repeat Steps 2–5 predicting $price$ instead of $lprice$.